Machine Learning for teens

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Ways to Overcome Overfitting

- **Feature Selection**
  - Simplify the model by removing irrelevant or noisy features.
- Regularization
  - Regularization works for linear models.
- Cross-Validation
  - KFold, StratifiedKFold, cross_val_score
- **Model Constraints** (for Trees)
  - Like our code
- Dimensionality Reduction
  - Reduce feature space like PCA

Ways to Overcome Overfitting

- Dropout
  - For neural networks
- Data Augmentation (for images/text)
  - Increase training diversity.
    - Flip, rotate, crop images
    - Synonym replacement for text
- Ensemble Methods
  - Combine models to reduce variance.
    - Bagging (Random Forest), Boosting (XGBoost), Stacking

Ensemble Methods

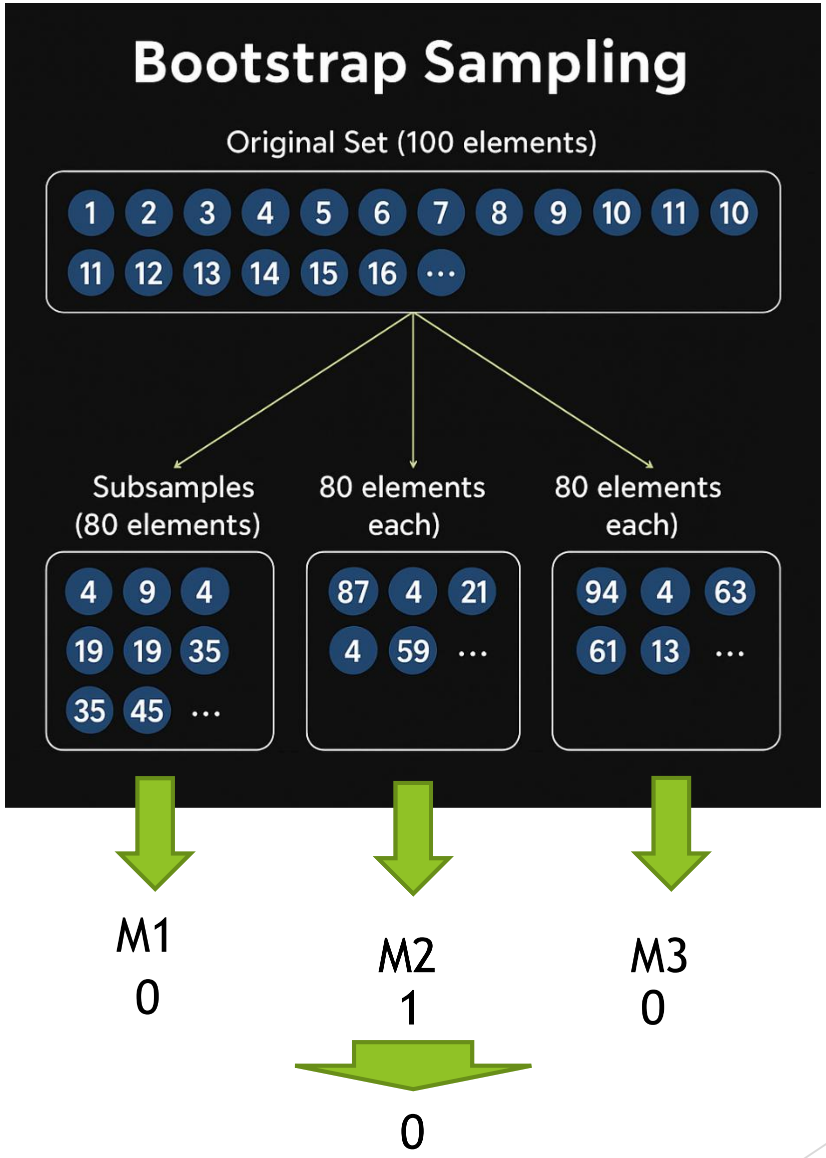
- Ensemble learning
  - Bagging(Bootstrap Aggregating)
  - Boosting
  - Stacking
- Bagging
  - is an ensemble method that trains multiple models independently on different random subsets of the data

How Bagging Works

- Bootstrap Sampling
  - Create multiple datasets by randomly sampling (with replacement) from the original training set
    - Each subset may contain duplicate samples
- Train Base Learners
  - Train a separate model (e.g., decision tree) on each bootstrap sample
    - Models are trained independently
      - Each learner is a weak leaner
- Aggregate Predictions
  - Classification
    - majority voting
  - Regression
    - averaging

Bagging

Bootstrap



Aggregation

Random forest

- Sample data
  - Features
  - Rows

No scaling the features

- **do not need to scale our features** when using **Bagging with Decision Trees**

Model selection in bagging

- Basically, model can be anything. If we use
  - Tree
    - Bagged trees
      - No need to scale data
        - care about which feature to split on — not how big the numbers are.
  - Other models
    - Bagging

TruncatedSVD for dimensionality reduction

- TruncatedSVD = PCA for sparse data
  - Sparse data
    - a dataset where **most of the values are zero**
  - Dense data
    - a dataset where most values are non-zero
- Reduces high-dimensional one-hot features to fewer components
- Prevents overfitting in Bagging/trees

OOB

- The **OOB (Out-Of-Bag) score** is essentially a **performance metric**
- Range
  - 0.0 → model predicts everything wrong
  - 1.0 → model predicts everything correctly

Assignment

- Load heart.csv and Prepare the Data
- Train a Single Decision Tree
- Train a Bagging Classifier with Decision Trees
- Compare results for a Single Decision Tree and Bagging